Vaibhav Dikshit

Mumbai, Maharashtra

+91-6386618353 vaibhavdxt14@gmail.com LinkedIn GitHub

SUMMARY

B.Tech **Electronics and Communication Engineering** undergraduate with strong foundations in **Embedded Systems, Automotive Electronics, Digital Electronics, and Microcontrollers**. Hands-on experience with ESP32, RFID, Bluetooth communication, and hardware-software integration through academic and personal projects. Passionate about developing reliable automotive and IoT solutions using communication protocols, real-time programming, and microcontroller-based system design.

EDUCATION

Vellore Institute of Technology, Bhopal 2023 – 2027
B.Tech in Electronics and Communication Engineering **CGPA: 9.55/10**

KDMA International, Kanpur 2022 – 2023
Class XII (PCM) — **95%**

Gulmohar Public School, Kanpur 2020 – 2021
Class X — **95.5%**

PROJECTS

IoT-Based Vehicle Health Monitoring System GitHub
ESP32, Embedded C, Wi-Fi, MQTT, Proteus **2026**

- Designed and simulated an IoT-based vehicle health monitoring system integrating **3 sensors** using ESP32 and Proteus.
- Implemented threshold-based fault detection with wireless telemetry using **MQTT**, enabling real-time monitoring.
- Created a modular Embedded C firmware with a configurable **1-second sensor sampling interval** for real-time data acquisition and alert generation.

CAN Bus Automotive Communication Simulator GitHub
ESP32, MCP2515, Embedded C, CAN Protocol, Proteus **2026**

- Simulated a CAN Bus network interconnecting **2 Electronic Control Units (ECUs)** for reliable in-vehicle communication.
- Implemented CAN frame transmission, message arbitration, interrupt-driven communication, and error handling using the **MCP2515 CAN controller** at **500 kbps**.
- Validated communication integrity by simulating the exchange of **100+ CAN messages** through Proteus virtual logic analysis.

RFID-Based University ID System GitHub
ESP32, MFRC522, SPI, Python, SQLite **2025**

- Developed an ESP32-based RFID authentication system integrating attendance, library and cafeteria services.
- Interfaced the MFRC522 RFID module using the SPI communication protocol for secure UID-based identification.
- Built a Tkinter GUI with SQLite backend for record management and automated Excel export, reducing manual effort by **80%**.

Bluetooth-Controlled Smart Automation System GitHub
ESP32, Embedded C, UART, Android **2025**

- Designed an ESP32-based embedded automation system enabling real-time Bluetooth communication with an Android application.
- Implemented UART-based command processing for controlling GPIO peripherals through a mobile interface.
- Tested communication reliability and validated fault handling under multiple operational scenarios.

TECHNICAL SKILLS

Embedded Systems: Embedded C, ESP32, Arduino, GPIO, Timers, Interrupts, PWM, ADC
Communication Protocols: UART, SPI, I2C, CAN, Bluetooth
Electronics: Digital Electronics, Analog Electronics, Microcontrollers, VLSI Design & Verification
Simulation & EDA Tools: Proteus, Keil uVision, LTSpice, Logisim, Cadence Virtuoso, Quartus Prime, GTKWave
Programming: C, C++, Python, Java, MATLAB
Developer Tools: Git, GitHub, VS Code

CERTIFICATIONS

- **VLSI Design and Verification** — Ethnus
- **Introduction to FPGA Programming** — Univ. of Colorado
- Introduction to IoT — NPTEL
- Introduction to Programming in C — NPTEL
- MATLAB Fundamentals — MathWorks
- Fundamentals of Signal Processing — MathWorks
- Signal Processing with MATLAB — MathWorks
- Core Signal Processing Techniques — MathWorks

LEADERSHIP & EXTRACURRICULAR

- Appointed **Floor Representative**; coordinated student–administration communication.
- Represented district in Chess; active participant in Table Tennis and Badminton.
- Organized and officiated college-level Table Tennis tournaments (50+ participants).